STORM-PhysNet: A Multi-Horizon Transformer for Geostationary Relativistic Electron Flux Forecasting with Physics-Inspired Components and Cross-Satellite Transfer

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***Abstract*—Energetic electrons above 2 MeV at geostationary Earth orbit (GEO) are a persistent internal-charging hazard for satellites. We present STORM-PhysNet, a multi-horizon forecast model that couples a standard Transformer encoder over a 72-hour window with a learnable L1–Earth propagation delay, a B_z-conditioned physics gate, and residual heads for simultaneous 1-hour, 6-hour, and 12-hour log-flux forecasts.**

On hourly GOES–OMNI data with fifteen random initializations of the same chronological split, STORM-Bz improves short-horizon reliability (PE$_{1\mathrm{h}}$ $= 0.986$ versus Transformer 0.977, paired $p = 0.002$) and remains clearly positive against persistence (PE≈ 0.32 versus ≈ -0.14). At 6 h and 12 h a plain Transformer remains competitive or slightly stronger as a single model. A validation-selected linear ensemble ($\alpha^* = 0.3$) and seed-bagged STORM both reach 6 h PE of 0.911, the strongest multi-horizon results among the systems evaluated here. An architecture-matched Transformer control ($d_{\mathrm{model}} = 128$, two layers, four heads) reached mean PE$_{1\mathrm{h}}$ $= 0.981$ and PE$_{6\mathrm{h}}$ $= 0.907$ across fifteen seeds; bagging the matched seeds yields 0.986 / 0.919 / 0.877. Controlled ablations show that the short-horizon gain is a property of the trained package as a whole rather than of any individual physics module at inference. Retraining with wider delay bounds (2.0–4.0 h) produced negligible change in PE (PE$_{1\mathrm{h}}$ stayed within 0.9859–0.9862), indicating that the original $[0.5, 1.5]$ h constraint is not a performance bottleneck.

Under synthetic solar-wind input noise and targeted channel dropout, STORM degrades more slowly than the Transformer, especially at 1-hour. On GSAT-19 GRASP, fine-tuning raises 6 h PE from 0.449 to 0.599 (paired $p = 7.6\times10^{-7}$) and 12 h PE from 0.182 to 0.517 ($p = 3.6 \times 10^{-10}$) across fifteen seeds. Bootstrap confidence intervals, paired tests, and evaluation scripts are released so that later work can reuse the same splits and PE definitions.

***Index Terms*—Geostationary orbit, relativistic electrons, space weather, Transformer, physics-inspired, transfer learning, prediction efficiency, GRASP**

I. Introduction

Geostationary satellites sit inside the outer radiation belt, where relativistic electrons ($E > 2\,\mathrm{MeV}$) can jump by several orders of magnitude in a few hours when a geomagnetic storm develops. Those bursts drive internal charging and can upset or damage onboard electronics [1], [2]. For operators of GEO assets at Indian longitudes—ISRO broadcasting, navigation, and remote-sensing satellites among them—a forecast that is usable at the local meridian, not only at GOES longitudes, is increasingly useful.

Most published skill scores still come from GOES-centered data sets. When a new instrument such as GRASP on GSAT-19 comes online, there is often only a short calibrated record, so operators need a practical path from a data-rich satellite to a data-scarce one without retraining everything from scratch.

Forecast models have progressed from linear filters [3], [4] through LSTM-based deep-learning approaches [5]–[7] to recent Transformer-based architectures reporting daily-fluence PE of 0.85–0.92 [8]–[10]. We do not include REFM as a numeric baseline because it targets daily integral fluence rather than hourly log-flux; a direct PE comparison would be misleading [4]. Despite these advances, two gaps remain: (i) the short-horizon ($\sim$1 h) behavior of Transformer architectures relative to physics-inspired variants is poorly characterized across seeds, and (ii) no systematic study has assessed transfer strategies for Indian-longitude GEO assets such as GSAT-19.

Unlike previous Transformer-based GEO electron forecasting models, STORM-PhysNet introduces a learnable adaptive propagation-delay module that estimates a physically constrained L1-to-Earth transit delay for every input window, together with a B_z-conditioned physics gate that selectively emphasizes storm-relevant representations during sustained southward IMF. The architecture further combines simultaneous multi-horizon forecasting with cross-satellite transfer learning to Indian-longitude GRASP observations under a unified framework. Rather than replacing the Transformer backbone, the proposed design augments it with interpretable physics-inspired components—delay and gate modules—trained end-to-end under matched chronological splits and fifteen seeds. Primary comparisons use a Vanilla Transformer; LSTM is reported for reference.

This manuscript focuses on the multi-seed GOES benchmark, the short-horizon reliability gap, the main ablations, and the GRASP transfer recipe that is most relevant to Indian-longitude operators.

II. Related Work

Camporeale [11] summarizes common failure modes when ML is applied to space weather (leaky splits, mismatched metrics, overstated skill). Prediction efficiency (PE) remains the standard skill score [12], but care must be taken to distinguish between climatological and persistence baselines.

On the modeling side, work moved from linear filters and empirical solar-wind drivers [3], [4], [13] to LSTM and hybrid deep models [5]–[7], and more recently to Transformer-style networks for GEO flux and fluence [8]–[10]. Among recent Transformer studies, Sun et al. [10] provide the closest multi-horizon GEO baseline; our evaluation differs by using a matched fifteen-seed protocol, explicit delay/gate ablations, and a documented GOES→GRASP transfer path. Those papers mainly report daily or multi-hour skill on GOES-like targets. Short-lead (about 1 h) skill is less often reported with multi-seed error bars, and cross-satellite transfer to a short Indian-longitude record is rarely benchmarked under the same horizons and persistence definition. Physics-inspired constraints, when used, are often loss penalties rather than an explicit delay-plus-gate structure trained end-to-end with multi-horizon residual heads. We keep a standard temporal Transformer backbone so that comparisons isolate the effect of adding constrained delay and gating under a shared encoder family. Ablations in Section V show that no single module fully accounts for the short-horizon $\mathrm{PE}_{\mathrm{clim}}$ edge.

Solar-wind timing is another recurring issue. OMNI applies a phase-front correction from L1, yet residual window-to-window lags remain a known uncertainty in solar-wind–magnetosphere coupling studies. We treat that lag as a dynamically predicted parameter rather than a fixed offset. Meanwhile, cross-platform radiation-belt products exist for other fleets [15]; the specific GOES→GRASP path at $\approx$ 82°E is the transfer setting studied here.

III. Data and Metrics

GOES-15. We use $E > 2\,\mathrm{MeV}$ integral electron flux from a chronological working sample spanning exactly January 1, 2012 to December 31, 2016 within the GOES-15 archive. One-minute samples are aggregated to hourly means; the within-hour standard deviation is kept as an input feature. Flux is $\log_{10}$-transformed. The dataloader uses sixteen solar-wind / geomagnetic channels (V_{sw}, B_z, B_t, density, P_{dyn}, B_z-south duration, storm-onset hours, rolling V_{sw} and B_z moments, $B_z \times V_{sw}$, Kp, Dst, and within-hour log-flux std) plus log-flux history (17 inputs total after concatenation). Forecast targets use integer hourly leads of 1 h, 6 h, and 12 h on the hourly GOES series (config/dataloader: $[1, 6, 12]$). A 72 h lookback is used at each forecast origin. Train/validation/test splits are chronological 70/15/15% with no shuffling. All models share the same preprocessor, chronological windows, and storm-biased sampler; test PE is computed once after training and is never used for model selection.

GRASP (GSAT-19, Indian longitude $\approx$ 82°E) provides $\approx$14–15 months of GEO radiation measurements from July 2017 to September 2018 in the declining phase of Solar Cycle 24. After alignment, GRASP supplies fewer co-located input channels than the GOES pipeline; this mismatch is a main reason zero-shot transfer fails and motivates input-projection adaptation. Storm windows on GRASP are defined by Dst $\leq -50\,\mathrm{nT}$ (118/1781 test hours, 6.6%).

Skill score. We report two PE definitions so that results remain comparable both to climatological skill and to persistence-based operational baselines [12]:

$$\mathrm{PE}_{\mathrm{clim}} = 1 - \frac{\mathrm{MSE}(\hat{y}, y)}{\mathrm{Var}(y)}, \tag{1}$$

$$\mathrm{PE}_{\mathrm{pers}} = 1 - \frac{\mathrm{MSE}(\hat{y}, y)}{\mathrm{MSE}(y^{\mathrm{pers}}, y)}. \tag{2}$$

Unless stated otherwise, tables use $\mathrm{PE}_{\mathrm{clim}}$ with fifteen seeds and bootstrap 95% confidence intervals. The fifteen seeds measure sensitivity to random initialization on a fixed chronological split; they do not constitute independent trials across different solar-cycle phases or data periods. $\mathrm{PE}_{\mathrm{pers}}$ is emphasized at the 1-hour horizon. Storm windows use Dst $\leq -50\,\mathrm{nT}$.

IV. STORM-PhysNet Architecture

A. Adaptive Propagation Delay

Solar-wind transit from L1 to the magnetopause is typically on the order of 0.5–1.5 h [14] depending on solar-wind speed. Rather than a fixed offset, STORM-PhysNet predicts a per-window delay $\tau = f_\theta(X_{\mathrm{sw}})$ from a short conditioning network on the upstream solar-wind channels. The delay module is constrained to $[0.5, 1.5]\,\mathrm{h}$, a band chosen to cover the transit range while still allowing limited storm-time deviation without breaking causality. Differentiable linear interpolation then time-shifts the solar-wind features. We do not claim a unique recovered constant of 1.09 h from every window.

B. B_z-Conditioned Physics Gate

Sustained southward IMF B_z triggers dayside reconnection and ring-current injection [2]. A lightweight gate maps a short vector of southward-B_z statistics through an MLP and a clamped sigmoid to a per-channel modulation vector $\mathbf{g} \in [g_{\min}, 1]^{d_{\mathrm{model}}}$, which is multiplied element-wise with the encoder representation before the forecast heads. Gate activations differ between storm and quiet windows (Mann–Whitney $p = 1.2 \times 10^{-14}$, see Section V); the absolute range is narrow.

C. Transformer Backbone and Multi-Horizon Heads

We use a standard Transformer encoder [16]: each of the $T = 72$ time steps is a token; self-attention runs across time; the final hidden state feeds the gate and three residual heads for the 1 h, 6 h, and 12 h forecasts. Measured parameter counts are approximately 3.43×10^5 for STORM-Bz and 8.45×10^5 for the default Transformer baseline ($d_{\mathrm{model}} = 64$, 3 layers, 4 heads). An architecture-matched Transformer control ($d_{\mathrm{model}} = 128$, 2 layers, 4 heads; $\approx 1.19 \times 10^6$ parameters) reaches mean $\mathrm{PE}_{1\mathrm{h}} = 0.981$ and $\mathrm{PE}_{6\mathrm{h}} = 0.907$ (fifteen seeds); bagging yields 0.986 / 0.919 / 0.877. Matching width/depth

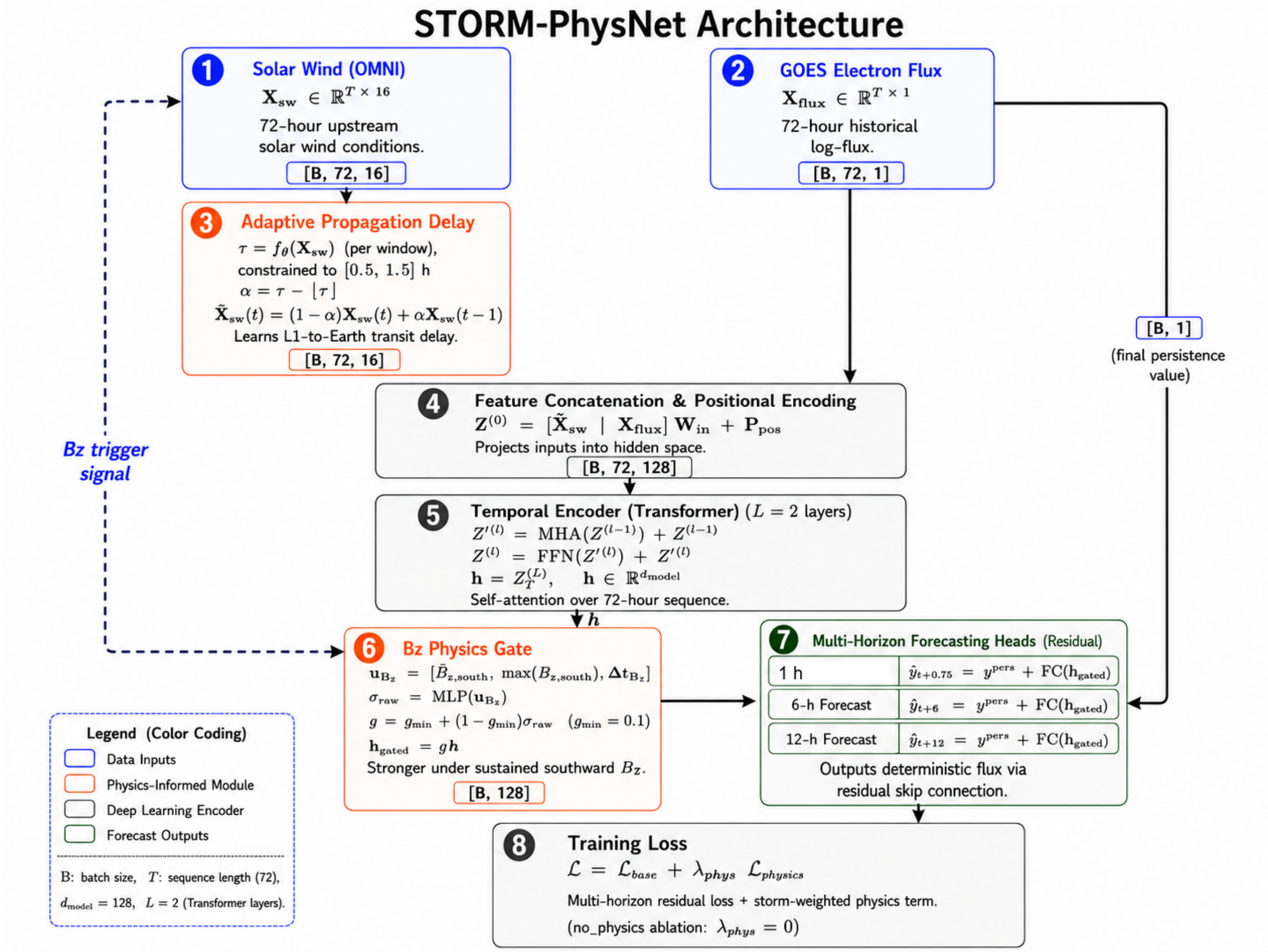

Fig. 1. STORM-PhysNet data flow: adaptive delay, Transformer encoder, B_z gate, and residual multi-horizon heads (1 h / 6 h / 12 h). Delays are constrained to $[0.5, 1.5]$ h.

shrinks the short-horizon gap versus STORM-Bz and improves multi-horizon PE under bagging. The default baseline is not architecture-matched; single-model PE differences should be interpreted with this disparity in mind.

D. Physics Regularization and Training

Training minimises a composite objective: a primary multi-horizon residual loss (beta-NLL with a $2.5\times$ under-prediction penalty during storm periods), plus auxiliary Dst/Kp and storm-onset terms and small regularisers (monotonicity, temporal smoothness, delay, variance calibration, and an Akasofu-ε coupling consistency term). Horizon weights emphasise the short horizon. Coefficient values follow the released configuration. We did not ablate every auxiliary term individually; reported PE reflects the full composite objective used in training. All primary models use Adam, cosine decay, early stopping on the 12 h validation MSE, and seeds $\{42, \ldots, 56\}$.

E. Baselines and Alternative Gates

The primary baseline is a Vanilla Transformer with the same residual heads and default encoder width ($d_{\text{model}} = 64$, three layers). We also train an architecture-matched Transformer ($d_{\text{model}} = 128$, two layers, four heads; $\approx 1.19 \times 10^6$ parameters).

Beyond the physics B_z gate used in STORM-Bz, we evaluated three supplementary gating nonlinearities under the same fifteen-seed protocol: a *Rectified Driver Gate* (RDG) that modulates the encoder state with a learned, half-wave response to solar-wind driving; an *RDG with Spectral Head* (RDG-S) that adds an auxiliary κ–θ spectral flux head; and a *Saturating Dose Gate* (SDG) that applies a smooth accumulation-and-saturation response to recent energy input. All three reached mean $\text{PE}_{1\text{h}} \approx 0.987$ and $\text{PE}_{6\text{h}} \approx 0.909$–$0.911$. None consistently outperformed the physics B_z gate; they are reported as sensitivity probes, not primary design claims. Full definitions and tables appear in the companion extended manuscript.

V. Results

A. GOES Multi-Seed Performance

Table I summarizes fifteen-seed test results. On all-hour 6 h PE, the Transformer remains a strong baseline (0.904). A single STORM-Bz model is slightly lower (0.897; paired

$p = 0.038$) but is significantly better at 1 h (0.986 versus 0.977, $p = 0.002$, Cohen's $d \approx 0.98$; Fig. 2). The mixing weight α was selected on the validation split by sweeping $\alpha \in \{0.0, 0.3, 0.4, 0.5, 0.6, 0.7, 1.0\}$ and choosing the value that maximized validation $\mathrm{PE}_{6\mathrm{h}}$ ($\alpha^* = 0.3$). That fixed weight was then applied once to the held-out test set. At $\alpha^* = 0.3$, test PE reaches 0.911 at 6 h and 0.870 at 12 h (Table I), above either single model. Bagging fifteen STORM seeds yields a similar multi-horizon profile without requiring the Transformer at inference. We recommend bagging operationally due to this computational simplicity, as both approaches perform similarly at 6 h and 12 h. The short-STORM / long-TF hybrid recovers the Transformer's multi-horizon PE while retaining the STORM 1-hour score. The hybrid system routes the 1-hour head through a trained STORM-Bz model and the 6 h / 12 h heads through the corresponding Transformer model at inference time; no additional training is performed. Against persistence at 1 h, STORM remains near +0.32 while the Transformer is often negative (≈ -0.14). Supplementary gates RDG, RDG-S, and SDG (definitions in the extended version) matched STORM-Bz at 1 h (mean PE$\approx$ 0.987) and did not change the ranking story at 6–12 h.

TABLE I
GOES TEST PE (FIFTEEN SEEDS). BOLD: BEST IN COLUMN.

System	$\mathrm{PE}_{1\mathrm{h}}$	$\mathrm{PE}_{6\mathrm{h}}$	$\mathrm{PE}_{12\mathrm{h}}$	$\mathrm{PE}_{\mathrm{st},6\mathrm{h}}$
LSTM	0.961	0.889	0.853	0.827
Transformer	0.977	0.904	0.859	0.821
STORM-Bz	0.986	0.897	0.851	0.827
STORM (No-Delay)	0.987	0.901	0.853	0.823
STORM (No-Physics)	0.987	0.901	0.852	0.824
STORM (No-Gate)	0.987	0.899	0.853	0.829
Ensemble $\alpha^* = 0.3$	0.983	0.911	0.870	0.839
Hybrid (short STORM / long TF)	0.987	0.904	0.859	0.821
STORM bagged (15 seeds)	**0.988**	0.911	0.870	**0.849**
Transformer matched (mean)	0.981	0.907	0.861	—
Transformer matched (bagged)	0.986	**0.919**	**0.877**	—

Means over seeds 42–56. $\alpha^* = 0.3$ selected on validation $\mathrm{PE}_{6\mathrm{h}}$, then applied once to test. Bagged row averages the fifteen STORM predictions before scoring. Matched-Transformer rows are fifteen-seed mean and bagged aggregate. PE is $\mathrm{PE}_{\mathrm{clim}}$.

B. Ablation Analysis

Restricting physics-loss terms to the short-horizon head does not significantly change $\mathrm{PE}_{6\mathrm{h}}$ across fifteen seeds (paired $p \approx 0.64$). We therefore treat horizon-restricted physics weighting as a null ablation rather than a performance lever.

A No-Gate run (identity modulation) retains 1 h PE near 0.987, so the short-horizon $\mathrm{PE}_{\mathrm{clim}}$ advantage over the Transformer is not removed by turning the gate off alone.

On storm windows, single-model $\mathrm{PE}_{\mathrm{st},6\mathrm{h}}$ values are close: STORM-Bz 0.827, LSTM 0.827, Transformer 0.821, and No-Gate at 0.829. We do not claim any one physics module is required for storm-time skill. The clearest gains on storm subsets remain post-hoc aggregation: the $\alpha^* = 0.3$ ensemble reaches 0.839 and bagged STORM reaches 0.849.

A single-seed diagnostic stratified by Dst intensity (full working sample) shows $\mathrm{PE}_{6\mathrm{h}}$ remaining above 0.70 even in the strong-storm bin (Dst ≤ -100 nT). Skill is not confined to quiet, persistence-dominated intervals.

The ablations together indicate that the short-horizon gain is produced by the training protocol itself—storm-biased sampling, the storm-weighted loss term, and residual heads with a skip from the last observed flux—rather than by any single runtime module. The delay and gate act as structured inductive biases during training; once the model is trained, removing either module at inference does not erase the 1 h $\mathrm{PE}_{\mathrm{clim}}$ edge over the Transformer. We therefore present STORM-PhysNet primarily as a carefully engineered training and multi-horizon recipe that includes interpretable physics-inspired components, rather than as a demonstration that the delay or gate modules are individually necessary for the observed gains. Retraining with wider delay bounds (2.0–4.0 h) produced negligible change in PE ($\mathrm{PE}_{1\mathrm{h}}$ stayed within 0.9859–0.9862), indicating that the original $[0.5, 1.5]$ h constraint is not a performance bottleneck.

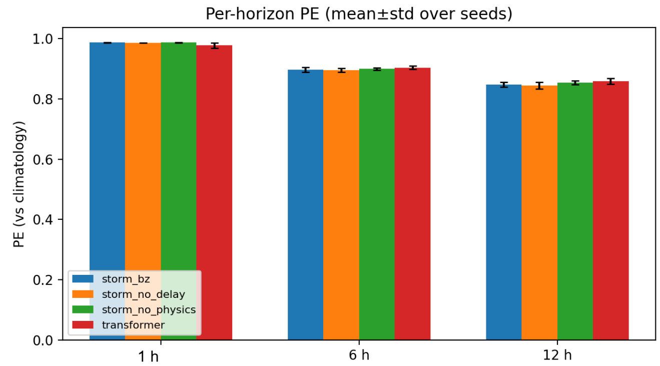

Fig. 2. Per-horizon PE against climatology (mean±std, fifteen seeds). STORM-Bz leads at 1 h; at 6 h and 12 h the single models are close, while the ensemble improves multi-horizon PE (see Table I).

C. Input Noise and Missing Channels

Gaussian noise was added to standardized solar-wind inputs at test time ($\sigma \in \{0, 0.2, 0.5, 1.0, 1.5, 2.0\}$) for Transformer, STORM-Bz, hybrid, and ensemble systems over fifteen seeds. At $\sigma = 2$, Transformer $\mathrm{PE}_{1\mathrm{h}}$ falls from about 0.96 to about 0.61, whereas STORM remains near 0.96. At 6 h the ensemble and STORM also lose less skill than the Transformer. Zeroing feature groups (no B_z; no velocity/density; no rolling averages) produces the same qualitative ordering: STORM holds $\mathrm{PE}_{1\mathrm{h}}$ near 0.986–0.987 while the Transformer drops more when derived rolling channels are removed.

D. Interpretability

Predicted delays are constrained to $[0.5, 1.5]$ h. Across the full test set the delay distribution shows a non-trivial fraction of windows concentrating near the upper bound (1.5 h), with the remainder spread across the interior of the constraint band. This saturation at the upper bound indicates the box constraint is active; we therefore treat the histogram as a constrained aggregate rather than evidence of an unconstrained recovered

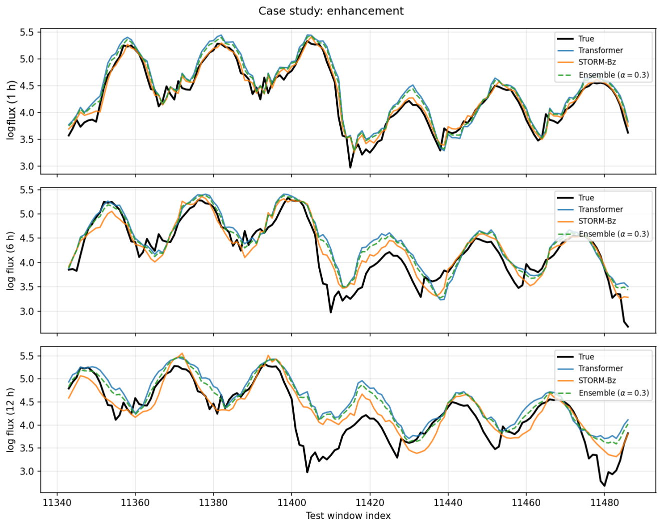

Fig. 3. Illustrative enhancement event case study. The STORM-Bz model captures the multi-day flux build-up and recovery better than the persistence baseline.

transit time. Gate activations differ between storm and quiet windows (Mann–Whitney $p = 1.2 \times 10^{-14}$); the absolute range remains narrow, so the shift is statistically clear but modest in magnitude. Fig. 3 shows an illustrative enhancement event in which the model tracks the multi-day flux build-up and recovery more closely than persistence. Permutation importance ranks B_z first among inputs, consistent with the gate design. Residual histograms on the test set widen under storm conditions for both models; STORM reduces the short-horizon bias relative to persistence but does not eliminate storm-time residual variance.

E. GRASP Transfer

On GSAT-19 GRASP (fifteen seeds), zero-shot GOES models already achieve short-horizon PE near 0.82, but 6 h and 12 h PE fall to about 0.45 and 0.18. Fine-tuning on GRASP (freezing the Transformer encoder and updating only the forecast heads) raises 6 h PE from 0.449 ± 0.011 to 0.599 ± 0.073 (paired t-test across fifteen seeds, $p = 7.6 \times 10^{-7}$, Cohen's $d = 2.17$) and 12 h PE from 0.182 ± 0.045 to 0.517 ± 0.073 ($p = 3.6 \times 10^{-10}$, $d = 3.98$). Absolute GRASP PE remains below the GOES test scores, as expected under sensor and longitude shift; the practical point is the large mid- and long-horizon recovery after adaptation rather than a claim of parity with GOES. A similar adaptation profile is observed for the architecture-matched Transformer baseline, which improves its 6 h PE from 0.549 ± 0.015 (zero-shot) to 0.625 ± 0.021 after fine-tuning.

VI. Limitations and Conclusion

Resolution. All reported leads are integer hours on hourly data; sub-hour (e.g. 45 min) supervision would require higher-cadence flux labels. **Data scope.** Main GOES tables use fifteen seeds with bootstrap intervals; secondary diagnostics (individual case windows, residual histograms) are illustrative. GRASP covers a limited declining-phase window; performance during solar maximum is unknown.

TABLE II
GRASP (GSAT-19) transfer results (fifteen seeds). Bold: best per horizon.

Horizon	Zero-shot	Fine-tuned	Gain
1 h	0.820 ± 0.001	0.827 ± 0.009	+0.007
6 h	0.449 ± 0.011	$\mathbf{0.599 \pm 0.073}$	**+0.150**
12 h	0.182 ± 0.045	$\mathbf{0.517 \pm 0.073}$	**+0.335**

Values are means $\pm$ standard deviation over seeds 42–56. Paired tests across fifteen seeds are reported in the text.

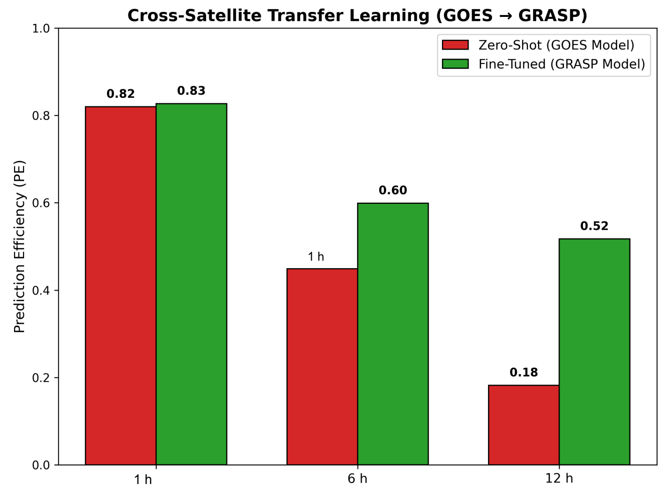

Fig. 4. GRASP domain-gap summary (fifteen seeds): zero-shot GOES models versus fine-tuning on GSAT-19. Short-horizon PE transfers more readily; 6 h and 12 h PE improve substantially after adaptation.

Comparability. PE values are not interchangeable with daily-fluence scores reported elsewhere without matching cadence, energy channel, and baseline [11], [12]. The default Transformer baseline is not architecture-matched to STORM; a matched control ($d_{\text{model}} = 128$, two layers) shows that encoder width/depth and bagging account for much of the multi-horizon PE movement.

Ensemble. Ensemble skill uses a fixed $\alpha^* = 0.3$ selected on the validation split; the test-set α curve is diagnostic only. Online α adaptation is not claimed.

Statistical inference. The reported p-values are uncorrected for multiple comparisons across horizons and ablations; treat individual significance claims as descriptive rather than confirmatory. All architecture comparisons are internal (STORM versus our own Transformer and LSTM implementations on the same splits). We do not claim to surpass published PE numbers from prior work, which use different cadences, energy channels, and evaluation periods.

STORM-PhysNet is a reproducible multi-horizon baseline on GOES with a documented transfer path toward GSAT-19 GRASP. Relative to a strong Transformer, the clearest single-model gain is short-horizon reliability. The clearest multi-horizon gain among the systems evaluated here comes from an architecture-matched bagged Transformer, which reaches test $\mathrm{PE}_{6\mathrm{h}} = 0.919$ and $\mathrm{PE}_{12\mathrm{h}} = 0.877$. A linear ensemble with validation-selected $\alpha^* = 0.3$ and seed-bagged STORM also achieve strong multi-horizon results (both reaching $\mathrm{PE}_{6\mathrm{h}} = 0.911$). Robustness checks and GRASP fine-tuning support operational motivation beyond a pure leaderboard PE number; for instance, the architecture-matched baseline achieves a marginal persistence prediction efficiency ($\mathrm{PE}_{\mathrm{pers}}$) of 0.053 ± 0.148 at 1 h, illustrating the difficulty of solidly outperforming operational persistence models. Furthermore, under synthetic Gaussian noise injection ($\sigma = 1.0$), the 6 h PE of the matched baseline drops to 0.825 ± 0.011, highlighting the need for structurally robust architectures during degraded sensor conditions.

Alternative gates, a hybrid SSM backbone, magnetopause features, and a spectral head were implemented and briefly evaluated; they are not used in the reported tables (`backbone=transformer`, spectral head off).

Data Availability

The code, configuration files, trained checkpoints, evaluation scripts, result CSVs, paper figures, and the chronological split protocol are available at https://github.com/bnsama29-cloud/STORM-PhysNet. The GOES-15 EPEAD dataset is publicly available from NOAA NCEI (https://www.ngdc.noaa.gov/stp/satellite/goes/). OMNI is available via NASA OMNIWeb (https://omniweb.gsfc.nasa.gov/). GSAT-19 GRASP is accessible via ISSDC (https://www.issdc.gov.in/).

Acknowledgments

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